

Chapter 13. Existence, Uniqueness and Stability

of Stochastic nonlinear differential equations,
with piecewise constant arguments (SEPCA).

(SEPCA)

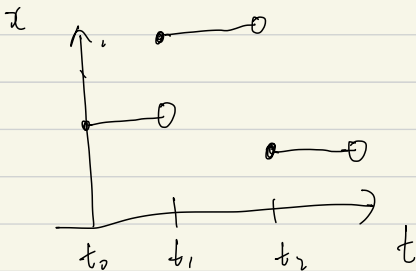
$$dx(t) = f(t, x(t), \lambda_{p(t)}(x(r(t)))) dt \quad \text{Ch 12}$$

$$+ g(t, x(t), \lambda_{p(t)}(x(r(t)))) dW(t)$$

$$x(t_0) = x_0,$$

$p(k)$: piecewise
 $r(k)$: constant. $k = \{k\}_{k=0}^{\infty}$
 $\tau = \{t_k\}_{k=0}^{\infty}$ in $t \in [t_k, t_{k+1})$

$$E_x, \quad t \in [t_0, t_1) \Rightarrow p(t) = 0 \\ r(t) = t_0.$$



$$\lambda_0(x(t_0)) \quad \lambda_2(x(t_2)) \\ \lambda_1(x(t_1))$$

For $t \in [t_k, t_{k+1})$, we can write

$$\begin{aligned} x(t) = & x^0 + \int_{t_0}^t f(s, x(s), \lambda_k(x(t_k))) ds \\ & + \int_{t_0}^t g(s, x(s), \lambda_k(x(t_k))) dW(s) \end{aligned} \quad (13.3)$$

1. Existence. (Linear growth of f, g)

2. Comparison method

3. Stability analysis.

Def. 13.1. A stochastic process $x, (a, b) \rightarrow \mathbb{R}$

solution of (SEPCA) if

i) $x(t)$ conti, F_t -adapted, $\forall t \in (a, b)$

(Def 2.26
in p32)

ii) $f(t, x(t), \lambda_k(x(t_k))) \in L_{ad}(\Omega, L^1(a, b))$

$g(t, x(t), \lambda_k(x(t_k))) \in L_{ad}(\Omega, L^2(a, b))$

F_t -adapted.
sample path.

L^p integrable.

iii) (13.3) holds, w.p. 1.

Def. 13.3, F_t -adapted, integrable process $X(t)$ is martingale, w.r.t filtration $\{F_t\}_{t \geq 0}$ if.

$$E[X(t) | F_s] = X(s)$$

conditional expectation of next future value is equal to present value.

Doob's martingale inequality,

Let $X(t)$: martingale., $p > 1$, $X(t) \in L^p(\Omega, \mathbb{R}^n)$

$$E \left[\sup_{a \leq t \leq b} \|X(t)\|^p \right] \leq \left(\frac{p}{p-1} \right)^p E \left[\|X(b)\|^p \right]$$

Borel - Cantelli's lemma.

If $\{A_k\}_{k=1}^{\infty} \subset F$ and $\sum_{k=1}^{\infty} P(A_k) < \infty$, then.

$$P \left(\limsup_{k \rightarrow \infty} A_k \right) = 0$$

13.1. Existence & Uniqueness of solution.

Thm 13.1 Assume.

(A1) f, g grow linearly :

$$\|f(t, x, y)\|^2 + \|g(t, x, y)\|^2 \leq L_1 (\|x\|^2 + \|y\|^2)$$

(A2) f, g : global Lipschitz :

$$\begin{aligned} & \|f(t, x_1, y_1) - f(t, x_2, y_2)\|^2 + \|g(t, x_1, y_1) - g(t, x_2, y_2)\|^2 \\ & \leq L_2 (\|x_1 - x_2\|^2 + \|y_1 - y_2\|^2), \end{aligned}$$

Then (SEPCA) has unique solution for $t \geq t_0$.

Proof).

$$(\text{Step 1}) \quad \max_{1 \leq n \leq j} E[\|x_n(t)\|^2] \leq C_2 e^{3L_1(t_1 - t_0 + 1)(t_1 - t_0)}$$

(we will define Cauchy sequence) for $t \in [t_0, t_1]$

$$(\text{Step 2}) \quad E[\|x_{n+1}(t) - x_n(t)\|^2] \leq \frac{C[F(t - t_0)]^n}{n!}, \quad (\text{f.h. to be determined})$$

$$(\text{Step 3}), \quad \lim_{n \rightarrow \infty} x_n(t) = x(t)$$

$$\text{Note. } a_n = \sqrt{n} \Rightarrow a_{n+1} - a_n = \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \rightarrow 0,$$

but $\lim_{n \rightarrow \infty} a_n \rightarrow \infty$

For $t \in [t_0, t_1)$, we define.

$$x_0(t) = x^0$$

$$\left\{ \begin{aligned} x_n(t) &= x^0 + \int_{t_0}^t f(s, x_{n-1}(s), \lambda_0(x_{n-1}(t_0))) ds \\ &\quad + \int_{t_0}^t g(s, x_{n-1}(s), \lambda_0(x_{n-1}(t_0))) dW(s) \\ x_n(t_0) &= x^0. \end{aligned} \right.$$

$$E[\|x_n(t)\|^2] \leq 3 E[\|x^0\|^2]$$

$$+ 3 E\left[\left|\int_{t_0}^t f(s, \dots) ds\right|^2\right]$$

$$+ 3 E\left[\left|\int_{t_0}^t g(s, \dots) dW(s)\right|^2\right]$$

$$\leq 3 E[\|x^0\|^2]$$

$$+ 3(t_1 - t_0) E\left[\int_{t_0}^t |f(s, \dots)|^2 ds\right] \quad \text{Holder in } t.$$

$$+ 3 \int_{t_0}^t E[|g(s, \dots)|^2] dt. \quad \text{Itô isometry}$$

(4g.2)

$$\left(\begin{aligned} &\leq 3 E[\|x^0\|^2] + 3(t_1 - t_0) \int_{t_0}^{t_1} \left[1 + E[\|x_{n-1}(t)\|^2] \right. \\ &\quad \left. E[\|\lambda_0(x_{n-1}(t_0))\|^2] \right] dt \\ &\quad + 3 \int_{t_0}^t \left[1 + E[\|x_{n-1}(t)\|^2] + E[\|\lambda_0(x_{n-1}(t_0))\|^2] \right] dt \end{aligned} \right)$$

$$\leq C_1 + 3L_1(t_1 - t_0 + 1) \int_{t_0}^{t_1} E[\|x_{n-1}(s)\|^2] ds.$$

$$\Rightarrow \max_{1 \leq n \leq j} E[\|x_n(t)\|^2] \leq C_2 + 3L_1(t_1 - t_0 + 1) \max_{1 \leq n \leq j} \int_{t_0}^{t_1} E[\|x_{n-1}(s)\|^2] ds.$$

By the Gronwall inequality

$$\max_{1 \leq n \leq j} E[\|x_n(t)\|^2] \leq C_2 e^{3L_1(t_1 - t_0 + 1)(t_1 - t_0)}$$

(Step 2) we use the induction argument.

when $n=0$.

$$\begin{aligned} E[\|x_1(t) - x_0(t)\|^2] &\leq 2 E\left[\left|\int_{t_0}^t f(s, x_0(s), \lambda_0(x_0(t_0))) ds\right|^2\right] \\ &\quad + 2 E\left[\left|\int_{t_0}^t g(s, x_0(s), \lambda_0(x_0(t_0))) dW(s)\right|^2\right] \\ &\leq 2L_1(t_1 - t_0)(1 + t_1 - t_0) \\ &\quad \times \left(1 + E[\|x^0\|^2] + E[\|\lambda_0(x^0)\|^2]\right) \\ &= C. \end{aligned}$$

Assume, $n \leq k$ holds.

when $n = k+1$,

$$\mathbb{E} \left[\underbrace{\|x_{n+2}(t) - x_{n+1}(t)\|^2}_{A_{n+2}} \right] \leq \underbrace{2L_2(t-t_0+1)}_{:= M_1}$$

$$\times \mathbb{E} \left[\int_{t_0}^t \left(\underbrace{\|x_{n+1}(s) - x_n(s)\|^2}_{A_{n+1}} + \underbrace{\|\lambda_0(x_{n+1}(t_0)) - \lambda_0(x_n(t_0))\|^2}_{=0} \right) ds \right]$$

$$A_{n+2}(t) \leq M \int_{t_0}^t A_{n+1}(s) ds$$

$$= M \int_{t_0}^t \frac{C [M(s-t_0)]^n}{n!} ds$$

$$= \frac{C [M(t-t_0)]^{n+1}}{(n+1)!}$$

(Step 3) (Step 2 implies)

$$\mathbb{E} \left(\sup_{t_0 \leq t < t_1} \|x_{n+1}(t) - x_n(t)\|^2 \right) \leq \frac{C M(t_1-t_0)^n}{n!}$$

$$\mathbb{P} \left\{ \sup_{t_0 \leq t < t_1} \|x_{n+1}(t) - x_n(t)\| > \frac{1}{2^n} \right\} \leq \frac{C M(t-t_0)^n}{n!}$$

By Borel-Cantelli's lemma.

$$\sup_{t_0 \leq t < t_1} \|x_{n+1}(t) - x_n(t)\| \leq \frac{1}{2^n}$$

$$x_n(t) = x_0(t) + \sum_{j=0}^{n-1} \underbrace{(x_{j+1}(t) - x_j(t))}_{\text{geometric sequence.}}$$

there exists limit point.

$\Rightarrow$

$$\lim_{n \rightarrow \infty} \|x_n(t) - x(t)\|_{L^2} \rightarrow 0$$

$$x(t) \stackrel{?}{=} x^0 + \int_{t_0}^t f(s, x(s), \lambda_0(x(t_0))) ds$$

$$\downarrow \text{in } L^2 \quad + \int_{t_0}^t g(s, x(s), \lambda_0(x(t_0))) dW(s)$$

$$x_{n+1}(t) = x^0 + \int_{t_0}^t f(s, x_n(s), \lambda_0(x_n(t_0))) ds$$

$$+ \int_{t_0}^t g(s, x_n(s), \lambda_0(x_n(t_0))) dW(s)$$

$$RHS \leq L_2(t_1 - t_0 + 1) \int_{t_0}^{t_1} E[\|x(s) - x_n(s)\|^2] ds$$

13.2 Comparison method.

Thm 13.2 Assume

(i). for any $k = 0, 1, 2, \dots$, $V \in C^{1,2}([t_k, t_{k+1}) \times \mathbb{R}^n; \mathbb{R})$

V is bounded below &

$$\underline{V}(t, x) \leq \underline{h}(t, x, \sigma_k(x)) \quad \text{a.s. } t \in [t_k, t_{k+1})$$

where h : concave, non decreasing in x , σ_k ,
 σ_k : concave function

$$(ii) \quad \dot{u}(t) = \underline{h}(t, u(t), \sigma_k(u(t))) \quad t \in [t_k, t_{k+1})$$

$$u(t_0) = u_0$$

has maximal solution $\psi(t; t_0, u_0)$ for all $t \geq t_0$.

$$\Rightarrow E[V(t_0, x_0)] \leq u_0 \text{ implies } E[V(t, x)] \leq \psi(t; t_0, x_0)$$

13.3 Stability Analysis

Thm 13.3. Suppose the same assumption of Thm 13.2.

Let $\exists a, b \in K$ s.t

$$b(\|u\|^2) \leq V(u) \leq a(\|u\|^2),$$

Then stability property of $u=0$ imply

stability of $x=0$ of (SEPCA).